

NIFTY 100 FINANCIAL INTELLIGENCE PLATFORM

Comprehensive Analyst Guide & System Reference

Version 1.0 · Institutional Equity Research & Fundamental Analysis Division

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Section 1: Platform Overview & Architecture

The Nifty 100 Financial Intelligence Platform is a production-grade fundamental equity analytics system designed for research analysts.

It transforms raw financial statements across 92 constituents of the Nifty 100 into institutional-grade KPIs, valuation screens, and interactive dashboards.

Key Architectural Layers:

- Layer 1 (Ingestion): Reads raw Excel files with automatic header and formatting normalization.
- Layer 2 (ETL & DQ): Enforces 16 Data Quality rules preventing duplicate, invalid, or corrupted records.
- Layer 3 (Storage): SQLite relational database maintaining complete foreign-key integrity.
- Layer 4 (Analytics): Vectorized 50+ KPI computation engine including CAGR turnaround handling.
- Layer 5 (Intelligence): Capital allocation pattern matching, KMeans clustering, and automated NLP pros/cons.
- Layer 6 (Reporting): Fully automated PDF tearsheets, sector reports, and portfolio summaries.
- Layer 7 (Interface): Multi-screen Streamlit application and high-performance REST API.

Section 2: Data Quality & Normalization Standard

Data fidelity is guaranteed through 16 automated Data Quality rules executed at ingest time:

- DQ-01: Company Primary Key Uniqueness — Enforces unique NSE ticker identifiers.
- DQ-02: Annual Time-Series Uniqueness — Enforces unique (company_id, year) composite pairs.
- DQ-03: Foreign Key Integrity — Strictly rejects orphaned child records lacking master company entries.
- DQ-04: Balance Sheet Equality — Flags balance sheet discrepancies exceeding 1% asset tolerance.
- DQ-05: Operating Margin Validation — Cross-validates source OPM against (operating_profit / sales).
- DQ-06: Revenue Plausibility — Audits zero or negative revenues for non-financial companies.
- DQ-07: Year Standardisation — Coerces diverse fiscal formats (e.g. 'Mar 23', 'FY24') into 'YYYY-MM'.
- DQ-08: Ticker Normalization — Enforces uppercase formatting and validates ticker character patterns.

Section 3: Financial KPI Definitions & Formulas

Comprehensive formulas utilized across the analytics layer:

1. Return on Equity (ROE) = $\text{Net Profit} / (\text{Equity Capital} + \text{Reserves}) \times 100$. Evaluates return on net assets.
2. Return on Capital Employed (ROCE) = $\text{EBIT} / \text{Capital Employed} \times 100$. Measures total capital efficiency.
3. Net Profit Margin (NPM) = $\text{Net Profit} / \text{Sales} \times 100$. Expresses final post-tax profitability.
4. Debt-to-Equity (D/E) = $\text{Total Borrowings} / \text{Net Worth}$. Measures corporate financial leverage.
5. Interest Coverage Ratio (ICR) = $(\text{Operating Profit} + \text{Other Income}) / \text{Finance Costs}$.
6. Free Cash Flow (FCF) = $\text{Operating Cash Flow} + \text{Investing Cash Flow}$.
7. Cash Flow Quality Ratio = $\text{Cash from Operations} / \text{Net Profit}$. Identifies accrual lockup when $< 0.5x$.

Section 4: Compound Growth (CAGR) & Turnaround Logic

Standard CAGR calculations fail when companies experience turnaround from losses or transition to losses.

The Platform adopts a formal Decision Matrix:

- Base > 0 and End > 0: Evaluated standard CAGR = $((\text{End}/\text{Base})^{(1/n)} - 1) \times 100$.
- Base < 0 and End > 0: Turnaround State. CAGR is mathematically undefined; tagged as 'TURNAROUND ↑'.
- Base > 0 and End < 0: Deterioration to Loss. Tagged as 'DECLINE_TO_LOSS'.
- Base < 0 and End < 0: Consecutive Loss-making. Tagged as 'BOTH_NEGATIVE'.
- Base = 0: Zero Base denominator error. Tagged as 'ZERO_BASE'.
- History < n years: Tagged as 'INSUFFICIENT'.

Section 5: Capital Allocation Pattern Matrix

Every company-year is categorized into one of 8 distinct capital allocation models based on cash flow signs:

- (+, -, -): 'Reinvestor' or 'Shareholder Returns' — Cash generated from core operations is reinvested into capex while returning excess to shareholders.
- (+, -, +): 'Expansion via External Capital' — Operations, debt, and equity all finance massive growth.
- (+, +, -): 'Asset Divestment & Debt Repayment' — Monetizing assets to deleverage.
- (-, -, +): 'High Growth / External Funding' — Early-stage capital-intensive ramp.
- (-, -, -): 'Severe Cash Distress' — Burning operational cash, incurring capex, and repaying capital.
- (-, +, +): 'Asset Sale to Fund Operations' — Distress signal requiring immediate analyst triage.

Section 6: Investment Screener Usage & Presets

The Screener engine allows multi-parameter filtering across 18 fundamental metrics.

Six Built-in Presets:

1. Quality Compounder: ROE > 15%, D/E < 1.0, FCF > 0, Revenue CAGR 5yr > 10%.
2. Value Pick: P/E < 20, P/B < 3.0, D/E < 2.0, Dividend Yield > 1.0%.
3. Growth Accelerator: PAT CAGR 5yr > 20%, Revenue CAGR 5yr > 15%, D/E < 2.0.
4. Dividend Champion: Dividend Yield > 2.0%, Dividend Payout < 80%, FCF > 0.
5. Debt-Free Blue Chip: D/E = 0.0, ROE > 12.0%, Sales > ₹5,000 Cr.
6. Turnaround Watch: Revenue CAGR 3yr > 10.0%, FCF positive latest year, D/E < 2.5.

Section 7: Composite Health Scoring Model

The Composite Financial Quality Score (0–100) eliminates single-metric distortion:

- Profitability (35% Weight): ROE (15%), ROCE (10%), Net Profit Margin (10%).
- Cash Generation Quality (30% Weight): FCF Compounding (15%), CFO/PAT (10%), Positive FCF (5%).
- Topline & Bottomline Growth (20% Weight): Revenue CAGR 5yr (10%), PAT CAGR 5yr (10%).
- Balance Sheet Solvency (15% Weight): D/E Ratio (10%), Interest Coverage (5%).

All underlying metrics undergo P10/P90 Winsorization to prevent extreme outliers from skewing results.

Section 8: Peer Comparison & Benchmark Engines

Covers 11 dedicated peer groups representing core sectors of the Indian economy.

Within each group, companies are evaluated across 20 distinct metrics using percentile ranking (0–100%).

- Best-in-Class Detection: Companies placing in the top quartile (≥ 75 th percentile) on ≥ 6 of 10 primary metrics.
- Watch List Detection: Companies placing in the bottom quartile (≤ 25 th percentile) on ≥ 4 of 10 metrics.
- Benchmark Gap Analysis: Direct metric-level spread vs group benchmark leader (e.g. TCS for IT, HDFCBANK for Private Banks).

Section 9: Valuation & Market Multiples Analysis

Utilizes historical and current valuation multiples to uncover mispricing opportunities:

- P/E vs 5-Year Historical Median: Identifies whether a company is expanding or compressing valuation.
- Sector-Relative P/E: Flags 'Caution (Overvalued)' if P/E > 1.5x sector median; flags 'Discount (Undervalued)' if < 0.7x sector median.
- Free Cash Flow Yield: Compares post-capex cash generation against equity market capitalization.

Section 10: Statistical Clustering & Portfolio Distribution

KMeans clustering (k=5) classifies the Nifty 100 universe into 5 behavioral archetypes:

1. High-Quality Growth: Exceptional returns on equity, robust margins, and zero-to-low debt.
2. Defensive Dividend: Stable cash cows with above-average dividend yield and moderate growth.
3. Value Cyclical: Capital-intensive commodity and industrial players trading at conservative multiples.
4. Emerging Growth: High reinvestment rates with rapid topline scaling.
5. Distressed / High Debt: Highly leveraged balance sheets or erratic operational earnings.

Section 11: Automated Reporting Layer

The platform generates complete publication-ready PDF documents:

- Company Tearsheets: 2-page detailed summaries including 10-year financials, charts, and qualitative insights.
- Sector Overviews: 11 sector publications summarizing constituent matrices and median benchmarks.
- Master Portfolio Summary: Unified catalog of all 92 Nifty 100 companies.

All reports adhere strictly to print margins, typography hierarchies, and automated vector chart embedding.